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A Proof of the Riemann Hypothesis via the Divisibility Property of Entire Functions

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Abstract This paper presents a proof of the Riemann Hypothesis (RH), in which the divisibility of entire functions arising from the symmetric functional equation, the uniqueness of zero multiplicities, and the Euler product property (which guarantees the divisibility of entire functions by confining all the zeros of the completed Riemann zeta function $\xi(s)$ to the critical strip) play pivotal roles. We first develop a new absolutely and uniformly convergent expression for $\xi(s)$ by pairing the conjugate zeros ρ_i and $\bar{\rho}_i$, counted with their multiplicities, in the Hadamard product, i.e.

$$\xi(s)=\xi(0)\prod_{\rho}\Big(1-\frac{s}{\rho}\Big)=\xi(0)\prod_{i=1}^{\infty}\Big(1-\frac{s}{\rho_i}\Big)\Big(1-\frac{s}{\bar{\rho}_i}\Big)=\xi(0)\prod_{i=1}^{\infty}\Big(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}+\frac{(s-\alpha_i)^2}{\alpha_i^2+\beta_i^2}\Big)^{m_i}$$

where ρ ranges over all zeros of $\xi(s)$, $(\rho_i,\bar{\rho}_i)$ range over all conjugate zero pairs of $\xi(s)$, $\rho_i=\alpha_i+j\beta_i$ with $0<\alpha_i<1$, $0<|\beta_1|\le|\beta_2|\le\cdots$, $\sum_{i=1}^{\infty}\frac{1}{\beta_i^2}<\infty$, and $m_i\ge 1$ is the multiplicity of the i-th quadruple of zeros $(\rho_i,\bar{\rho}_i,1-\rho_i,1-\bar{\rho}_i)$. Then, according to the symmetric functional equation $\xi(s)=\xi(1-s)$, we obtain

$$\prod_{i=1}^{\infty}\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big)^{m_i}=\prod_{i=1}^{\infty}\Big(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}\Big)^{m_i}$$

which, owing to the divisibility of entire functions, uniqueness of m_i, and the irreducibility of each real quadratic polynomial factor, is finally equivalent to

$$\alpha_i=\frac{1}{2}, i\in\mathbb{N}, 0<|\beta_1|<|\beta_2|<|\beta_3|<\cdots$$

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Thus, we conclude that the RH is true.

Keywords Riemann Hypothesis; Hadamard product; symmetric functional equation; divisibility property of entire functions; uniqueness of zero multiplicities; Euler product; irreducibility of real quadratic polynomial factor

Notation
$\mathbb{N}$: the set of natural numbers, i.e., $\mathbb{N} = \{1, 2, 3, \ldots\}$.
$\mathbb{R}$: the field of real numbers.
$\mathbb{C}$: the field of complex numbers (the complex plane).
$\mathbb{R}[x]$: the ring of real polynomials.
$\mathbb{H}(\mathbb{C})$: the ring of entire functions.

It should be noted that in this paper, 'j' is used to denote the imaginary unit ($j^2 = -1$), while 'i' serves as natural number index.

1 Introduction

The Riemann zeta function is originally defined in the half-plane $\Re(s) > 1$ [1] by the absolutely convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \Re(s) > 1 \tag{1}$$

The connection between the above-defined Riemann zeta function and prime numbers was discovered by Euler, i.e., the famous Euler product

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p} (1 - p^{-s})^{-1}, \Re(s) > 1 \tag{2}$$

where p runs over the prime numbers.
The absolute convergence of the Euler product for $\Re(s) > 1$ guarantees that $\zeta(s) \neq 0$ for $\Re(s) > 1$; see Ref. [2, p. 51] and Ref. [3, p. 102] for more details. This means that $\zeta(s)$ has no zeros with real part greater than 1. This fact will play an important role in what follows, where it will be combined with the symmetric functional equation and other facts to confine all non-trivial zeros of $\zeta(s)$ to the critical strip.

Riemann showed in his paper in 1859 how to extend the zeta function to the whole complex plane $\mathbb{C}$ by analytic continuation

$$\zeta(s) = \frac{\Gamma(1 - s)}{2\pi j} \int_{\infty}^{\infty} \frac{(-x)^s}{e^x - 1} \cdot \frac{dx}{x} \tag{3}$$

where " $\int_{\infty}^{\infty}$ " is the symbol adopted by Riemann to represent the contour integral from $+\infty$ to $+\infty$ around a domain which includes the value 0 but no other point of discontinuity of the integrand in its interior, 'j' is the imaginary unit ($j^2 = -1$).

Or equivalently,

$$\zeta(s) = \frac{\pi^{s/2}}{\Gamma(s/2)}\{\frac{1}{s(s-1)} + \int_1^{\infty}(x^{\frac{s}{2}-1} + x^{-\frac{s}{2}-\frac{1}{2}}) \cdot (\frac{\theta(x)-1}{2})dx\} \quad (4)$$

where $\theta(x) = \sum_{-\infty}^{\infty} e^{-n^2\pi x}$ is the Jaccobi theta function, $\Gamma(s)$ is the Gamma function in the following Weierstrass expression

$$\frac{1}{\Gamma(s)} = s \cdot e^{\gamma s}\prod_{n=1}^{\infty}(1+\frac{s}{n})e^{-s/n} \quad (5)$$

where γ is the Euler-Mascheroni constant.

As shown by Riemann, $\zeta(s)$ extends to $\mathbb{C}$ as a meromorphic function with only a simple pole at $s = 1$, with residue 1, and satisfies the following functional equation

$$\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\zeta(s) = \pi^{-\frac{1-s}{2}}\Gamma(\frac{1-s}{2})\zeta(1-s) \quad (6)$$

The Riemann zeta function $\zeta(s)$ has zeros at the negative even integers: -2, -4, -6, -8, $\cdots$ and one refers to them as the **trivial zeros**. The other zeros of $\zeta(s)$ are complex numbers, i.e., **non-trivial zeros**.

In 1896, Hadamard [4] and Poussin [5] independently proved that no zeros could lie on the line $\Re(s) = 1$, together with the functional equation $\xi(s) = \xi(1-s)$ and the fact that there are no zeros with real part greater than 1, this showed that all non-trivial zeros must lie in the interior of the **critical strip** $0 < \Re(s) < 1$. Later on, Hardy (1914) [6], Hardy and Littlewood (1921) [7] showed that there are infinitely many zeros on the **critical line** $\Re(s) = \frac{1}{2}$.

To give a summary of the related research works on the RH, we have the following results on the properties of the non-trivial zeros of $\zeta(s)$ [8,p.160].

Lemma 1: Non-trivial zeroes of $\zeta(s)$, noted as $\rho = \alpha + j\beta$, have the following properties

1) The number of non-trivial zeroes is infinity;
2) $\beta \neq 0$;
3) $0 < \alpha < 1$;
4) $\rho, \bar{\rho}, 1-\bar{\rho}, 1-\rho$ are all non-trivial zeroes.

As further study, the completed zeta function $\xi(s)$ is proposed, i.e.

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\zeta(s) \quad (7)$$

It is well-known that $\xi(s)$ is an entire function of order 1. In addition, replacing s with $1-s$ in Eq.(7), and combining Eq.(6), we obtain the following functional equation

$$\xi(s) = \xi(1-s) \quad (8)$$

According to the definition of $\xi(s)$ in Eq.(7), and recalling Eq.(5), the trivial zeros of $\zeta(s)$ are canceled by the poles of $\Gamma(\frac{s}{2})$[9,p.217]. The zeros of $s(s-1)$ cancel the simple pole of $\zeta(s)$ at $s = 1$, and the pole $s = 0$ of $\Gamma(\frac{s}{2})$,

respectively [10,p.260]. Thus, all the zeros of $\xi(s)$ are exactly the nontrivial zeros of $\zeta(s)$. Then we have the following Lemma 2.

Lemma 2: The zeros of $\xi(s)$ coincide with the non-trivial zeros of $\zeta(s)$. Consequently, the following two statements are equivalent.

Statement 1: All the non-trivial zeros of $\zeta(s)$ have real part equal to $\frac{1}{2}$.

Statement 2: All zeros of $\xi(s)$ have real part equal to $\frac{1}{2}$.

The starting point of this paper is the Hadamard product expansion of $\xi(s)$, which was first proposed by Riemann and later rigorously justified by Hadamard [11]

$$\xi(s) = \xi(0) \prod_{\rho} (1 - \frac{s}{\rho}), \quad s \in \mathbb{C} \tag{9}$$

where $\xi(0) = \frac{1}{2}$, ρ runs over all zeros of $\xi(s)$.

Hadamard showed that to ensure the absolute convergence of this infinite product expansion, ρ and $1 - \rho$ must be paired. Later in Section 3, we will demonstrate that pairing ρ with $\bar{\rho}$ (together with the boundedness of α_i guaranteed by the Euler product property) also ensures the absolute and uniform convergence of the infinite product on compact subsets of $\mathbb{C}$.

2 Preliminary Lemmas

Lemmas 3 - 6 are provided to support the proof of the RH in the next section. We begin with the ring of real polynomials $\mathbb{R}[x]$, defined as

$$\mathbb{R}[x] = \{\sum_{i=0}^{\infty} a_i x^i | a_i \in \mathbb{R}, a_i \neq 0 \text{ for all but a finite number of } i\}$$

and equipped with the operations + (addition) and · (multiplication).

The ring of real polynomials is a subset of the ring of entire functions, which is defined as the set of all holomorphic functions on the whole complex plane $\mathbb{C}$, together with the operations of addition and multiplication, denoted as $\mathbb{H}(\mathbb{C})$, see Ref. [12, p. 197] and Ref. [13] for more details.

Both rings possess properties of divisibility, coprimality, and the greatest common divisor, denoted as "gcd". There are also differences between these two rings. Among others, the polynomial ring is a unique factorization domain (UFD), while the ring of entire functions is not a UFD. For entire functions, their divisibility, coprimality and common factors are determined by the relationships between their zero sets, see Ref. [14, p. 174] for more details.

Lemma 3: Let $f(s)$ be an entire function on $\mathbb{C}$ with $f(0) \neq 0$. Suppose $f(s) = \prod_{i=1}^{\infty} g_i(s)$ converges absolutely and uniformly on compact subsets of $\mathbb{C}$, where each $g_i(s) \in \mathbb{R}[s]$ is irreducible of degree $d \in \{1, 2\}$. If $m(s) \in \mathbb{R}[s]$ is irreducible of degree d and divides $f(s)$, then $m(s)$ divides $g_i(s)$ for some $i \geq 1$.

Proof: Let α be a root of $m(s)$, i.e., $m(\alpha) = 0$. Since $m(s) \mid f(s)$, we have $f(\alpha) = 0$. By absolute convergence of $\prod_{i=1}^{\infty} g_i(s)$ at $s = \alpha$, there exists at

least one index $i \in \mathbb{N}$ such that $g_i(\alpha) = 0$, otherwise $g_i(\alpha) \neq 0$ for all i, then $\prod_{i=1}^{\infty} g_i(\alpha)$ converges to a non-zero limit, contradicting $f(\alpha) = 0$.

As $g_i(s)$ and $m(s)$ are irreducible in $\mathbb{R}[s]$ with $\deg(g_i(s)) = \deg(m(s)) = d$, they share the root α. Thus:

- If $d = 1$, then $g_i(s) = a(s - \alpha)$ and $m(s) = b(s - \alpha)$ for $a, b \neq 0$, so $m(s) \mid g_i(s)$.
- If $d = 2$, then both $g_i(s)$ and $m(s)$ have roots $\{\alpha, \bar{\alpha}\}$, so $g_i(s) = c \cdot m(s)$ for $c \neq 0$, hence $m(s) \mid g_i(s)$.

In both cases, $m(s)$ divides $g_i(s)$.

That completes the proof of Lemma 3.

Remark: Although the uniform convergence is not used directly in the proof of Lemma 3, it guarantees (by Weierstrass's theorem) that the infinite product defines an entire function.

The following Lemma 4 and Lemma 5 help to understand the finiteness and uniqueness of the multiplicities of zeros of $\xi(s)$.

Lemma 4: Let $f(s)$ be a non-zero entire function, and let s_0 be a zero of $f(s)$. Then the multiplicity of s_0 is a finite positive integer.

Proof: Let $f(s) \not\equiv 0, s \in \mathbb{C}$, be an entire function, which means it is holomorphic on the whole complex plane. Suppose $f(s)$ has a zero at $s_0 \in \mathbb{C}$ of multiplicity m, then $f(s) = (s - s_0)^m g(s)$, where $g(s)$ is also an entire function and $g(s_0) \neq 0$.

Assume for contradiction that m is infinite, which implies there exists an accumulation point of zeros in the neighbor of s_0. Then, by Identity Theorem for holomorphic functions, and considering "0" is also an entire function, we have $f(s) \equiv 0, s \in \mathbb{C}$, which contradicts the given condition that $f(s) \not\equiv 0, s \in \mathbb{C}$. Thus, the assumption is false, i.e., m must be a finite positive integer.

That completes the proof of Lemma 4.

Remark: Statements similar to Lemma 4 can be found in Ref. [15, p. 44] and other related textbooks/monographs.

Lemma 5: Let $f(s)$ be a non-zero entire function, and let s_0 be a zero of $f(s)$. Then the multiplicity of s_0 is unique.

Proof: Let $f(s) \not\equiv 0, s \in \mathbb{C}$, be an entire function, which has a multiple zero at $s_0 \in \mathbb{C}$ of multiplicity m. We can write: $f(s) = (s - s_0)^m g(s)$, where $g(s)$ is also an entire function and $g(s_0) \neq 0$.

Assume for contradiction that there exists another integer $n \neq m$ such that n is also a multiplicity of the zero s_0. This means we can also write: $f(s) = (s - s_0)^n h(s)$, where $h(s)$ is an entire function and $h(s_0) \neq 0$.

Since both expressions for $f(s)$ must be equal, we then obtain $(s-s_0)^m g(s) = (s - s_0)^n h(s)$. Without loss of generality, consider $m > n$, then we have: $(s - s_0)^{m-n} g(s) = h(s) \Rightarrow h(s_0) = 0$, which is a contradiction to $h(s_0) \neq 0$.

Thus, the assumption is false, i.e., the multiplicity of a zero of any non-zero entire function is unique.

That completes the proof of Lemma 5.

Remark: Based on Lemma 4 and Lemma 5, the multiplicity m_i of the i-th quadruple of zeros $(\alpha_i + j\beta_i, \alpha_i - j\beta_i, 1-\alpha_i - j\beta_i, 1-\alpha_i + j\beta_i)$ of $\xi(s)$ is finite and uniquely determined.

Lemma 6: The infinite product $\prod_{i=1}^{\infty}\left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}\right)^{m_i}$ converges to a non-zero constant, given the conditions: $0<\alpha_i<1, 0<|\beta_1|\le|\beta_2|\le|\beta_3|,\cdots, \sum_{i=1}^{\infty}\frac{1}{\beta_i^2}<\infty$ (with each β_i^2 appearing m_i times), and $1\le m_i<\infty$ is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$ of $\xi(s)$.

Proof: Let $a_i = \frac{\alpha_i^2}{\alpha_i^2+\beta_i^2}$, then $\frac{\beta_i^2}{\alpha_i^2+\beta_i^2} = 1-a_i$. Since $0<\alpha_i<1$ and $\beta_i\neq 0$, we have $0<a_i<\frac{1}{\beta_i^2}$. By the given condition $\sum_{i=1}^{\infty}\frac{1}{\beta_i^2}<\infty$ (with each β_i^2 appearing m_i times), we have equivalently $\sum_{i=1}^{\infty}\frac{m_i}{\beta_i^2}<\infty$ (with $|\beta_1|<|\beta_2|<\cdots$. Hence $\sum_{i=1}^{\infty} m_i a_i < \sum_{i=1}^{\infty}\frac{m_i}{\beta_i^2}<\infty$, which guarantees that the product $\prod_{i=1}^{\infty}(1-a_i)^{m_i} = \prod_{i=1}^{\infty}\left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}\right)^{m_i}$ converges to a non-zero constant.

That completes the proof of Lemma 6.

Remark: In Lemma 6 and similar situations, the condition $\sum_{i=1}^{\infty}\frac{1}{\beta_i^2}<\infty$ with $0<|\beta_1|\le|\beta_2|\le\cdots$ (where each β_i^2 appears m_i times) is equivalent to $\sum_{i=1}^{\infty}\frac{m_i}{\beta_i^2}<\infty$ with $0<|\beta_1|<|\beta_2|<\cdots$, where m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$.

3 A Proof of the RH

This section presents a proof of the Riemann Hypothesis. We first prove that Statement 2 of the RH is true, and then by Lemma 2, Statement 1 of the RH is also true. To be brief, to prove the Riemann Hypothesis, it suffices to show that $\alpha_i = \frac{1}{2}, i\in\mathbb{N}$ in the new expression of $\xi(s)$ as shown in Eq.(13).

Proof of the RH: The proof proceeds in three steps.

Step 1:

It is well-known that zeros of $\xi(s)$ always come in complex conjugate pairs. Then by pairing $\rho_i = \alpha_i + j\beta_i$ and $\bar{\rho}_i = \alpha_i - j\beta_i$ in the Hadamard product as shown in Eq.(9), we have

$$\begin{aligned}\xi(s) &= \xi(0)\prod_{\rho}(1-\frac{s}{\rho}) = \xi(0)\prod_{i=1}^{\infty}(1-\frac{s}{\rho_i})(1-\frac{s}{\bar{\rho}_i})\\ &= \xi(0)\prod_{i=1}^{\infty}(1-\frac{s}{\alpha_i+j\beta_i})(1-\frac{s}{\alpha_i-j\beta_i}) = \xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}+\frac{(s-\alpha_i)^2}{\alpha_i^2+\beta_i^2})\end{aligned} \tag{10}$$

where $0 < \alpha_i < 1, \beta_i \neq 0$ (according to Lemma 1) are real numbers with $|\beta_1| \leq |\beta_2| \leq |\beta_3| \leq \cdots$, ρ ranges over all zeros of $\xi(s)$, $(\rho_i, \bar{\rho}_i)$ range over all conjugate zero pairs of $\xi(s)$.

The absolute convergence of the infinite product in Eq.(10) in the form

$$\xi(s) = \xi(0)\prod_{i=1}^{\infty}(1-\frac{s}{\rho_i})(1-\frac{s}{\bar{\rho}_i}) = \xi(0)\prod_{i=1}^{\infty}\Big(1-\frac{s(2\alpha_i - s)}{|\rho_i|^2}\Big) \tag{11}$$

depends on the convergence of infinite series $\sum_{i=1}^{\infty}\frac{1}{|\rho_i|^2}$ (since $|s| < \infty, 0 < \alpha_i < 1 \Rightarrow |s(2\alpha_i - s)| < \infty$), which is a well-known fact according to Theorem 2 in Section 2, Chapter IV of Ref. [2, p. 57] (also listed in Appendix for convenience). Thus, the infinite products as shown in Eq.(11) and Eq.(10) are absolutely convergent on the whole complex plane $\mathbb{C}$.

According to the Weierstrass M-test, these two infinite products as shown in Eq.(12) and Eq.(10) are also uniformly convergent on any compact subset of $\mathbb{C}$. Indeed, let $K \subset \mathbb{C}$ be an arbitrary compact set and let $R_K = \sup_{s\in K}|s|$. Then for all $s \in K$ and all i, since $0 < \alpha_i < 1$, we have

$$|s(2\alpha_i - s)| \leq |s|(2\alpha_i + |s|) < R_K(2 + R_K) = M_K.$$

Hence

$$\frac{|s(2\alpha_i - s)|}{|\rho_i|^2} < \frac{M_K}{|\rho_i|^2}, \qquad \text{and} \qquad \sum_{i=1}^{\infty}\frac{M_K}{|\rho_i|^2} = M_K\sum_{i=1}^{\infty}\frac{1}{|\rho_i|^2} < \infty.$$

Therefore, both infinite products as shown in Eq.(11) and Eq.(10) converge absolutely and uniformly on K.

Further, considering the absolute convergence of

$$\xi(s) = \xi(0)\prod_{i=1}^{\infty}\Big(1-\frac{s(2\alpha_i - s)}{|\rho_i|^2}\Big) = \xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2} + \frac{(s-\alpha_i)^2}{\alpha_i^2+\beta_i^2}) \tag{12}$$

we have the following new expression of $\xi(s)$ by putting all the possible multiple factors (zeros) together:

$$\xi(s) = \xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2} + \frac{(s-\alpha_i)^2}{\alpha_i^2+\beta_i^2})^{m_i} \tag{13}$$

where $m_i \geq 1$ is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$.

Step 2: Replacing s with $1 - s$ in Eq.(13), we obtain the infinite product expression of $\xi(1 - s)$, i.e.,

$$\xi(1-s) = \xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2} + \frac{(1-s-\alpha_i)^2}{\alpha_i^2+\beta_i^2})^{m_i} \tag{14}$$

According to the functional equation $\xi(s) = \xi(1-s)$, and considering Eq.(13) and Eq.(14), we have

$$\xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}+\frac{(s-\alpha_i)^2}{\alpha_i^2+\beta_i^2})^{m_i} = \xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}+\frac{(1-s-\alpha_i)^2}{\alpha_i^2+\beta_i^2})^{m_i} \quad (15)$$

which is equivalent to

$$\prod_{i=1}^{\infty}(1+\frac{(s-\alpha_i)^2}{\beta_i^2})^{m_i} = \prod_{i=1}^{\infty}(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2})^{m_i} \quad (16)$$

Since $0 < \alpha_i < 1$, then $\sum_{i=1}^{\infty}\frac{1}{|\rho_i|^2} < \infty \Rightarrow \frac{1}{|\rho_i|^2} \to 0 \Rightarrow |\rho_i|^2 \to \infty \Rightarrow |\beta_i|^2 \to \infty$.

Further we have $\lim_{i\to\infty}\frac{\beta_i^2}{|\rho_i|^2} = \lim_{i\to\infty}\frac{\beta_i^2}{\alpha_i^2+\beta_i^2} = 1$, that means $\sum_{i=1}^{\infty}\frac{1}{\beta_i^2}$ and $\sum_{i=1}^{\infty}\frac{1}{|\rho_i|^2}$ have the same convergence, i.e., $\sum_{i=1}^{\infty}\frac{1}{\beta_i^2} < \infty$. Thus, the infinite products on both sides of Eq.(16) are absolutely and uniformly convergent on compact subsets of $\mathbb{C}$, with reference to the analysis of Eqs. (10) and (11) in **Step 1**.

Furthermore, both sides of Eq.(16) converge to entire functions, because they differ with the entire functions $\xi(s)$ and $\xi(1-s)$ respectively by a non-zero multiplicative constant, i.e.

$$\begin{aligned}\xi(s) =& \xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}+\frac{(s-\alpha_i)^2}{\alpha_i^2+\beta_i^2})^{m_i}\\ =& \xi(0)\prod_{i=1}^{\infty}(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2})^{m_i}\prod_{i=1}^{\infty}(1+\frac{(s-\alpha_i)^2}{\beta_i^2})^{m_i}\\ =& c\cdot\prod_{i=1}^{\infty}(1+\frac{(s-\alpha_i)^2}{\beta_i^2})^{m_i}\end{aligned} \quad (17)$$

where c is a non-zero constant according to Lemma 6.

Step 3: Considering

$$\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)^{m_i} = \underbrace{\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}} \quad (18)$$

we have from Eq.(16) that

$$\prod_{i=1}^{\infty}\underbrace{\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}} = \prod_{i=1}^{\infty}\underbrace{\left(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}} \quad (19)$$

where the ith factor appears m_i times in any order.

Based on the conditions $0 < \alpha_i < 1$, $\sum_{i=1}^{\infty}\frac{1}{\beta_i^2} < \infty$ and the Weierstrass M-Test, both sides of Eq.(19) are infinite products absolutely and uniformly

convergent on compact subsets of $\mathbb{C}$. Hence each side of Eq.(19) can be rearranged into any quadratic polynomial times the remaining product, which also converges absolutely and uniformly on compact subsets of $\mathbb{C}$ (This reduces to the fact that omitting finitely many terms does not affect the convergence of an infinite product.), thus defines an entire function. Then by the definition of divisibility of entire functions [14,p.174], each quadratic factor on one side divides the product on the other side, i.e.,

$$\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big) \mid \prod_{i=1}^{\infty} \underbrace{\Big(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}\Big)\cdots}_{m_i \text{ times}} \tag{20}$$

$$\Big(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}\Big) \mid \prod_{i=1}^{\infty} \underbrace{\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big)\cdots}_{m_i \text{ times}} \tag{21}$$

where "|" is the divisible sign, $i \in \mathbb{N}$.

Both polynomials $1+\frac{(s-\alpha_i)^2}{\beta_i^2}$ and $1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}$ have discriminant $\Delta = -4\cdot\frac{1}{\beta_i^2} < 0$. Hence they are irreducible in $\mathbb{R}[s]$. By Lemma 3, Eq.(20) yields (Eq.(21) plays the same role as Eq.(20) and can be ignored):

$$\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big) \mid \Big(1+\frac{(1-s-\alpha_l)^2}{\beta_l^2}\Big), i,l \in \mathbb{N} \tag{22}$$

For the special kind of polynomials in Eq.(22), "divisible" means "equal", which can be verified by comparing the like terms in the following equation to get $k=1$.

$$\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big) = k\Big(1+\frac{(1-s-\alpha_l)^2}{\beta_l^2}\Big), k \neq 0 \in \mathbb{R} \tag{23}$$

Further, due to the uniqueness of the multiplicity m_i, the only solution to Eq.(22) is

$$\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big) \mid \Big(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}\Big), i \in \mathbb{N} \tag{24}$$

i.e.,

$$\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big) = \Big(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}\Big), i \in \mathbb{N} \tag{25}$$

since otherwise we have

$$\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big) \mid \Big(1+\frac{(1-s-\alpha_l)^2}{\beta_l^2}\Big), i,l \in \mathbb{N}, l \neq i \tag{26}$$

i.e.,

$$\Big(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\Big) = \Big(1+\frac{(1-s-\alpha_l)^2}{\beta_l^2}\Big), i,l \in \mathbb{N}, l \neq i \tag{27}$$

Eq.(27) leads to $\alpha_i + \alpha_l = 1, \beta_i^2 = \beta_l^2, l \neq i$, which means the i-th and the l-th quadruples of zeros coincide on the complex plane, violating the uniqueness of m_i.

Therefore, by comparing the like terms in Eq.(25), we obtain $\alpha_i = \frac{1}{2}, i \in \mathbb{N}$. Further, to ensure the uniqueness of m_i while $\alpha_i = \frac{1}{2}$, we need to restrict the β_i values to be distinct, i.e., $0 < |\beta_1| < |\beta_2| < |\beta_3| < \cdots$. Then we obtain

$$\alpha_i = \frac{1}{2}, i \in \mathbb{N}, 0 < |\beta_1| < |\beta_2| < \cdots \tag{28}$$

Thus, we conclude that all the zeros of the completed zeta function $\xi(s)$ have real part equal to $\frac{1}{2}$, i.e., Statement 2 of the RH is true. According to Lemma 2, Statement 1 of the RH is also true, i.e., all the non-trivial zeros of the Riemann zeta function $\zeta(s)$ have real part equal to $\frac{1}{2}$.
That completes the proof of the RH.

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Finally, with this manuscript, the author pays tribute to Bernhard Riemann and other predecessor mathematicians. They are the shining stars in the sky of human civilization.

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Appendix

Theorem 2.[2] The function $\xi(s)$ is an entire function of order one that has infinitely many zeros ρ_n such that $0 \leq \operatorname{Re}\rho_n \leq 1$. The series $\sum |\rho_n|^{-1}$ diverges, but the series $\sum |\rho_n|^{-1-\varepsilon}$ converges for any $\varepsilon > 0$. The zeros of $\xi(s)$ are the nontrivial zeros of $\zeta(s)$.